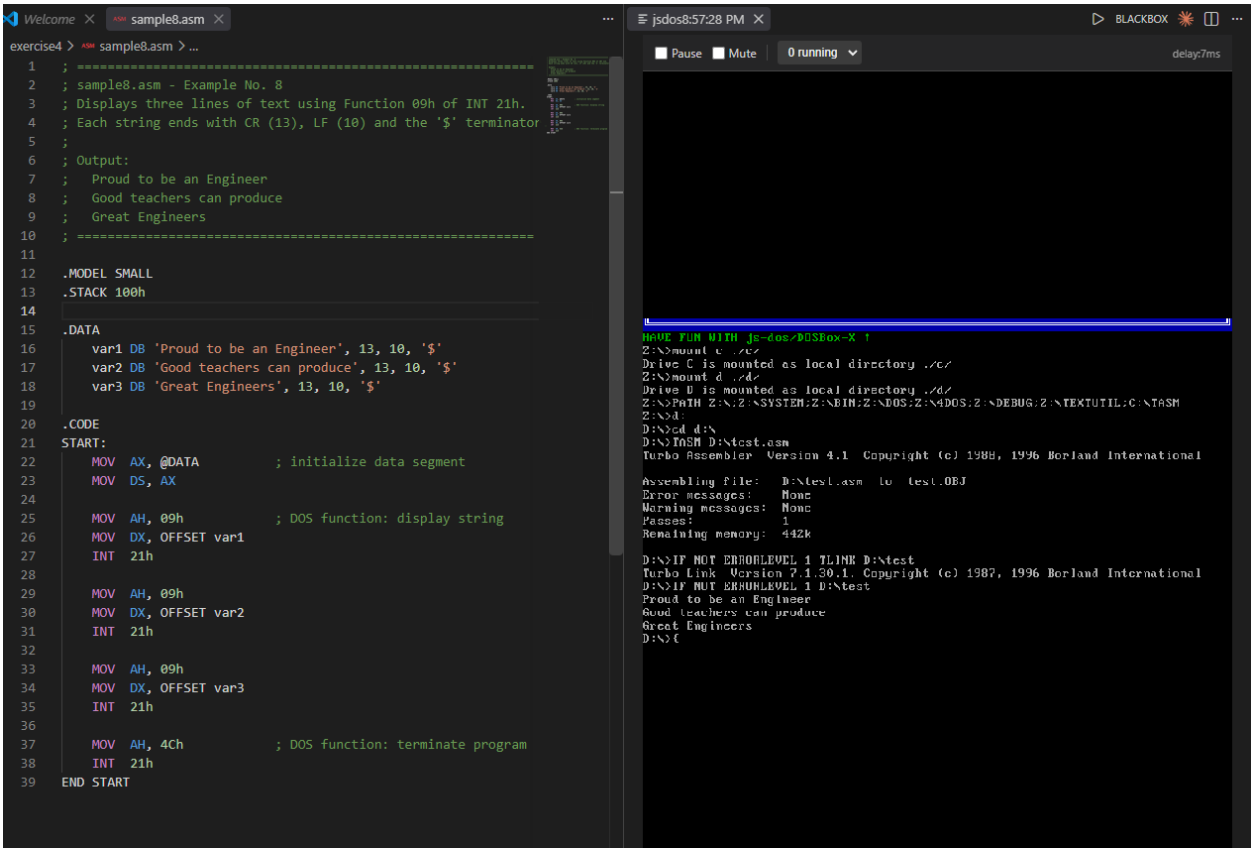
Latayan, Manuel R
CPE4A
CPE19L

Exercise 4

Example No. 1: The program will display the string characters shown below. Save the program using sample8.asm as the filename



```
; ============================================================
; sample8.asm - Example No. 8
; Displays three lines of text using Function 09h of INT 21h.
; Each string ends with CR (13), LF (10) and the '$' terminator
;
; Output:
;   Proud to be an Engineer
;   Good teachers can produce
;   Great Engineers
; ============================================================

.MODEL SMALL
.STACK 100h

.DATA
    var1 DB 'Proud to be an Engineer', 13, 10, '$'
    var2 DB 'Good teachers can produce', 13, 10, '$'
    var3 DB 'Great Engineers', 13, 10, '$'

.CODE
START:
    MOV  AX, @DATA          ; initialize data segment
    MOV  DS, AX

    MOV  AH, 09h            ; DOS function: display string
    MOV  DX, OFFSET var1
    INT  21h

    MOV  AH, 09h
    MOV  DX, OFFSET var2
    INT  21h

    MOV  AH, 09h
    MOV  DX, OFFSET var3
    INT  21h

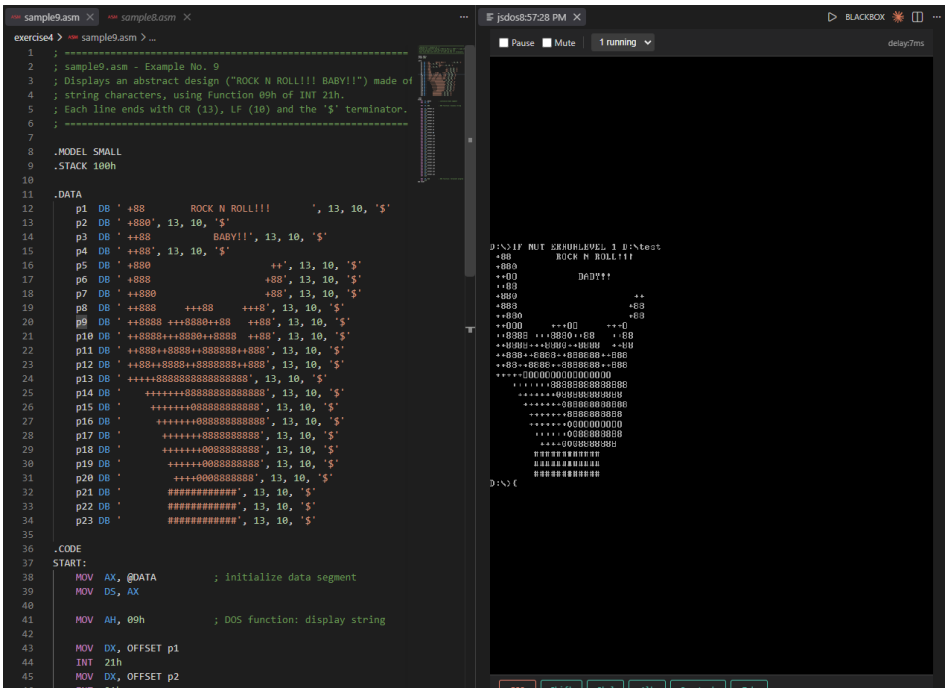
    MOV  AH, 4Ch            ; DOS function: terminate program
    INT  21h
END START
```

```
HAVE FUN WITH js-dos/DOSBox-X !
Z:\>mount c ./c/
Drive C is mounted as local directory ./c/
Z:\>mount d ./d/
Drive D is mounted as local directory ./d/
Z:\>PATH Z:\;Z:\SYSTEM;Z:\BIN;Z:\DOS;Z:\4DOS;Z:\DEBUG;Z:\TEXTUTIL;C:\TASM
Z:\>d:
D:\>cd d:\
D:\>TASM D:\test.asm
Turbo Assembler  Version 4.1  Copyright (c) 1988, 1996 Borland International

Assembling file:   D:\test.asm  to  test.OBJ
Error messages:    None
Warning messages:  None
Passes:            1
Remaining memory:  442k

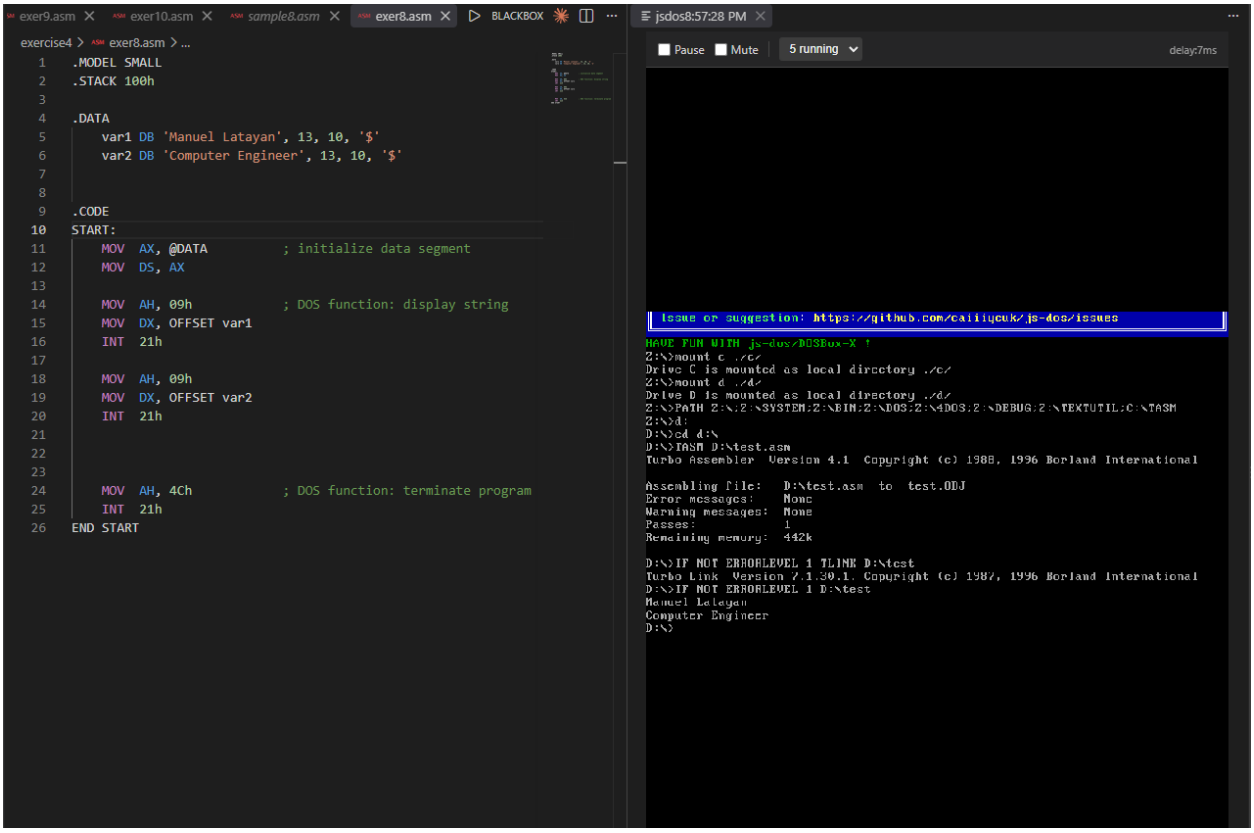
D:\>IF NOT ERRORLEVEL 1 TLINK D:\test
Turbo Link  Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
D:\>IF NOT ERRORLEVEL 1 D:\test
Proud to be an Engineer
Good teachers can produce
Great Engineers
D:\>
```

Example No. 2: Using the same function 09h and DOS INT 21h service routine, string characters showing abstract designs can also be created as shown below. Save the program using sample9.asm as the filename



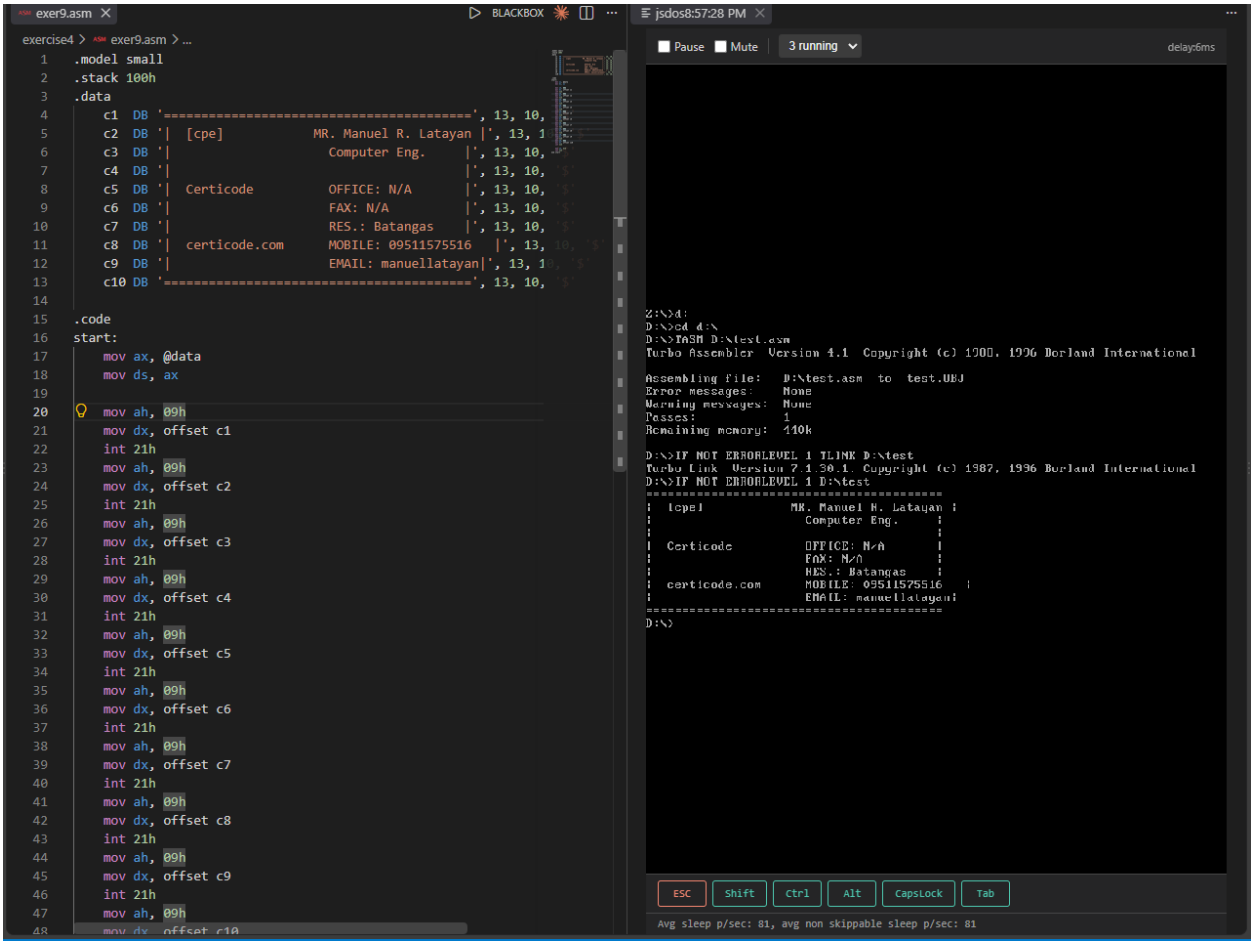
Machine Problem 1:

Create a program using Function 09h that will display the string output shown below using the carriage return and line feed option. Save the program using Exer8.asm as the filename.



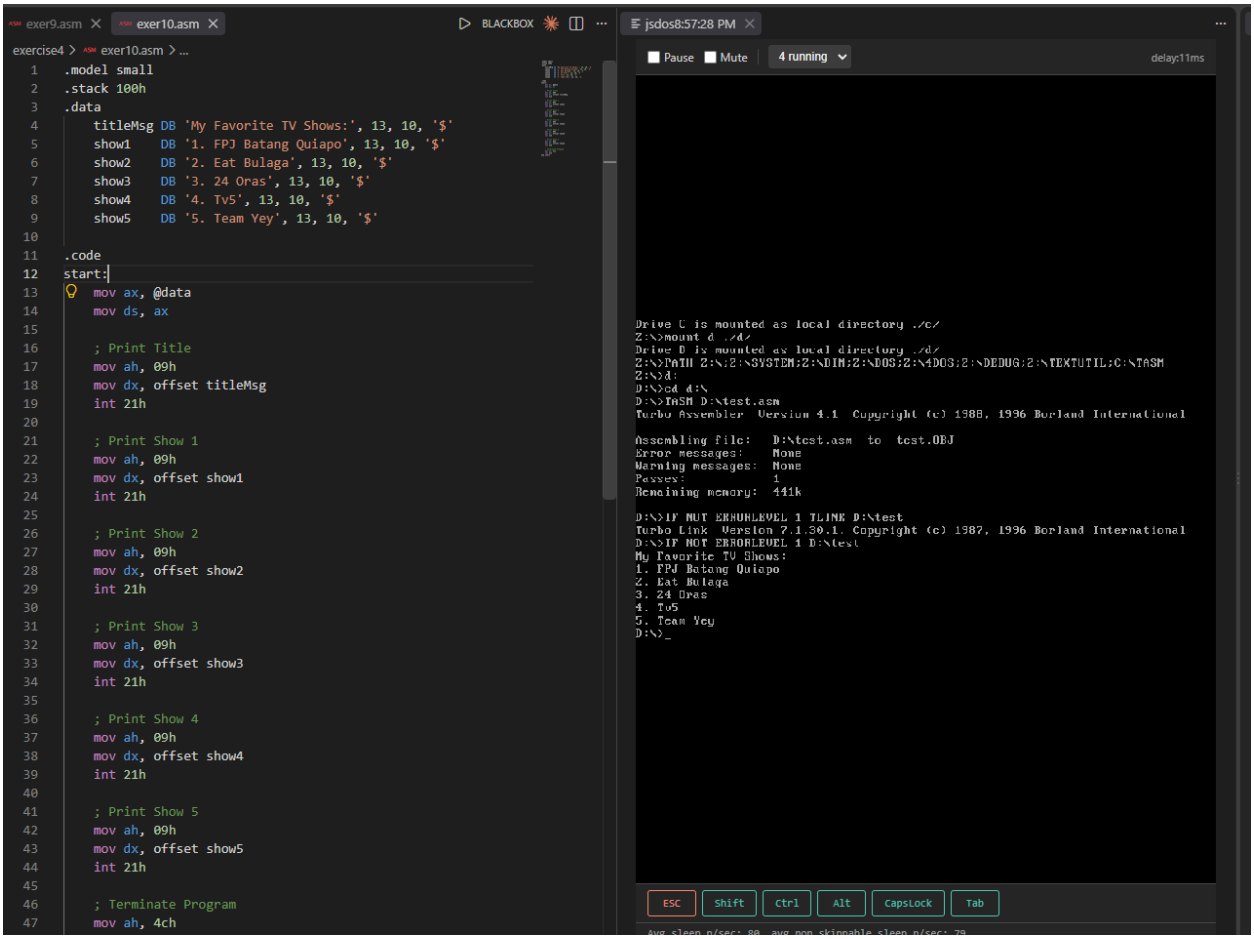
Machine Problem 2:

Create a personalized calling card that may follow the given format below using Function 09h that will display the string output. You can also choose your own border design and logo according to your design. Save the program using Exer9.asm as the filename.



Machine Problem 3:

Create a program using Function 09h that will display a list of your favorite TV shows in any TV Network local or international. Save the program using Exer10.asm as the filename



```
.model small
.stack 100h
.data
    titleMsg DB 'My Favorite TV Shows:', 13, 10, '$'
    show1    DB '1. FPJ Batang Quiapo', 13, 10, '$'
    show2    DB '2. Eat Bulaga', 13, 10, '$'
    show3    DB '3. 24 Oras', 13, 10, '$'
    show4    DB '4. Tv5', 13, 10, '$'
    show5    DB '5. Team Yey', 13, 10, '$'

.code
start:
    mov ax, @data
    mov ds, ax

    ; Print Title
    mov ah, 09h
    mov dx, offset titleMsg
    int 21h

    ; Print Show 1
    mov ah, 09h
    mov dx, offset show1
    int 21h

    ; Print Show 2
    mov ah, 09h
    mov dx, offset show2
    int 21h

    ; Print Show 3
    mov ah, 09h
    mov dx, offset show3
    int 21h

    ; Print Show 4
    mov ah, 09h
    mov dx, offset show4
    int 21h

    ; Print Show 5
    mov ah, 09h
    mov dx, offset show5
    int 21h

    ; Terminate Program
    mov ah, 4ch
```

```
Drive C is mounted as local directory ./c/
Z:\>mount d ./d/
Drive D is mounted as local directory ./d/
Z:\>PATH Z:\;Z:\SYSTEM;Z:\DIM;Z:\DOS;Z:\4DOS;Z:\DEBUG;Z:\TEXTUTIL;C:\TASM
Z:\>d:
D:\>cd d:\
D:\>TASM D:\test.asm
Turbo Assembler  Version 4.1  Copyright (c) 1988, 1996 Borland International

Assembling file:   D:\test.asm  to  test.OBJ
Error messages:    None
Warning messages:  None
Passes:            1
Remaining memory:  441k

D:\>IF NOT ERRORLEVEL 1 TLINK D:\test
Turbo Link  Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
D:\>IF NOT ERRORLEVEL 1 D:\test
My Favorite TV Shows:
1. FPJ Batang Quiapo
2. Eat Bulaga
3. 24 Oras
4. Tv5
5. Team Yey
D:\>_
```